

Photocatalytic reduction of CO₂ at (SnO₂, Fe₃O₄)/TiO₂ composite

Kamil Urbanek^{a, b}, Kaja Spilarewicz^a, Jiaguo Yu^c, Wojciech Macyk^{a, *}

^a Faculty of Chemistry, Jagiellonian University, ul. Gronostajowa 2, 30-387, Kraków, Poland

^b Doctoral School of Exact and Natural Sciences, Jagiellonian University, ul. Łojasiewicza 11, 30-348, Kraków, Poland

^c Laboratory of Solar Fuel, Faculty of Materials Science and Chemistry, China University of Geosciences, Wuhan, 430074, PR China



ARTICLE INFO

Article history:

Received 18 January 2023

Received in revised form

1 March 2023

Accepted 16 March 2023

Available online 21 March 2023

Keywords:

S-scheme

heterojunction

CO₂ reduction

spinel

ABSTRACT

In this work, the advanced photocatalytic system composed of titanium dioxide (P25) and mSFO (Sn-doped magnetite with Fe-doped cassiterite) was designed for the gas phase CO₂ reduction process. We have proposed the synthesis route of mSFO composite involving a clean inorganic hydrothermal method supported by the full characteristics of the obtained material by X-ray diffraction, X-ray photoelectron spectroscopy, X-ray fluorescence, Mössbauer spectroscopy, scanning electron microscopy, transmission electron microscopy, dynamic light scattering, diffuse reflectance spectroscopy, and porosimetry. The mSFO/P25 system indicates a photocatalytic reduction of CO₂ to CO. The activity of the examined photocatalytic system was also confirmed with the use of ¹³C labeled carbon dioxide. It was established that the formed carbon monoxide is a product of CO₂ reduction. The surface photovoltage, electrochemical (linear sweep voltammetry) and photoelectrochemical measurements substantiate the S-scheme mechanism.

© 2023 Elsevier Ltd. All rights reserved.

1. Introduction

Nowadays, the reduction of CO₂ is one of the most challenging research fields in photocatalysis among photocatalytic water splitting, CH₄ activation, nitrogen fixation, and photocatalytic fine chemicals synthesis. The main difficulty of this particular process is the consequence of the high thermodynamical stability of the CO₂ molecule ($\Delta G^0_{\text{formation}} = -394 \text{ kJ/mol}$) [1]. One electron CO₂ reduction process requires a potential of -1.9 V vs. normal hydrogen electrode (NHE) narrowing the list of potential photocatalysts to only a few since the photocatalytic reactions rely on the potentials of band edges of semiconductors. Multielectron processes, which do not require such high energy, exhibit low charge transfer rates preventing the efficient production of CO, oxygenates (HCOOH or CH₃OH), or hydrocarbons (CH₄, C_xH_{2x-2}). In addition to that, there are more concerns with the yields of these processes. The most unfavorable of them is the weak adsorption of CO₂ over the majority of inorganic materials. Moreover, many photocatalysts often require sacrificial hole acceptors to guarantee considerable efficiency. Therefore, a strong effort should be placed on the design of new materials, effective in photocatalytic CO₂ reduction.

Among photocatalysts, titania (TiO₂) is one of the most widely studied photocatalytic semiconductor materials. The photocatalytic activity of P25 (composite of titania phases with a typical content of 78% anatase, 14% rutile, and up to 8% amorphous phase) results from synergistic interactions between its components [2]. It became a benchmark for numerous photocatalytic systems [2]. However, this material is not an ideal solution for all applications, especially CO₂ reduction. Although the potential of the conduction band (CB) edge of TiO₂ is comparable to the potentials of CO₂ reduction (e.g., -0.24 V vs. NHE for reduction of CO₂ toward CH₄ in 8 electron process), a photocatalyst should offer significantly more negative CB edge energy (E_{CB}) to perform efficiently [3]. Therefore, titanium dioxide can be used in combination with other semiconductors (with better reducing properties) to enable photocatalytic CO₂ reduction. There are several possible ways of creating such hybrid systems, depending on the charge carrier separation mechanism: the conventional type-II heterojunction, a direct Z-scheme, and an S-scheme [4–8]. The strategy based on combining different semiconductors is highly promising also due to the efficient separation of the e^-/h^+ pairs, and consequently, their diminished recombination [9].

Several semiconductors exhibit E_{CB} more suitable for the photocatalytic reduction of carbon dioxide than P25. So far, the main efforts have been pushed into the investigation of such materials as g-C₃N₄ (-1.23 V vs. NHE) [10], CdS (-0.6 V vs. NHE) [11], ZnS

* Corresponding author.

E-mail address: macyk@chemia.uj.edu.pl (W. Macyk).

(−1.65 V vs. NHE) [12], and CuI (−2.28 V vs. NHE) [13], whereas an interesting and still insufficiently studied group of materials for application in photocatalytic reduction of CO₂ is spinel ferrites (MFe₂O₄, M = divalent metal ion) [14]. The main challenge in the application of this group of materials is the fact that their properties heavily depend on their phase, size, shape, and structure, thus many recent works focused on their controlled synthesis [15]. Additionally, the electronic properties of spinel ferrites can be tuned, and therefore, photocatalytic activity enhanced by doping. For example, Guo et al. presented that doping of ZnFe₂O₄ with Ce accelerates the production rates of H₂, CO, and CH₄ [16]. Another premise of the potential for the application of iron materials in the design of systems for CO₂ photocatalytic reduction is the high affinity of iron-bearing minerals, especially Fe₃O₄, to adsorption of CO₂ molecules at (001) and (111) surfaces [17].

Furthermore, SnO₂ is an interesting candidate to design the photocatalytic system for CO₂ reduction, due to its low CB position [18]. The most outstanding feature of this material is, however, its high electron mobility (*ca.* 100–200 cm²/V/S) [19]. Moreover, it presents high chemical stability and relatively easy synthesis routes resulting in stable nanoparticles [20]. Currently, limited data indicate that SnO₂ nanoparticles can exhibit photocatalytic activity toward CO₂ reduction. For example, Chowdhury et al. reported the photocatalytic reduction of CO₂ to HCOOH in the water phase using mesoporous SnO₂ nanoparticles under white LED light [21]. Moreover, Torres et al. showed that SnO₂ nanoparticles decorated with surface –OH groups are prone to photocatalytically reduce CO₂ to CH₄, CO, CH₃OH, and C₂H₄ in CO₂/water atmosphere [20]. The increase in photocatalytic activity of SnO₂ can be improved by the introduction of impurities by doping, which can reduce the incident photon energy and inhibit the recombination of e[−]/h⁺ pairs [22].

Herein, we propose for the first time the photocatalytic system composed of TiO₂ (P25) and composite (mSFO) of Sn-doped magnetite (Fe₃O₄) with Fe-doped cassiterite (SnO₂) for the application in photocatalytic reduction of CO₂, with water as a hole scavenger instead of other typically used sacrificial agents.

2. Experimental

2.1. Materials

Commercial analytical grade reagents: SnCl₂·2H₂O, FeCl₃·6H₂O, and NaOH were used in the synthesis of materials. The reagents were used as obtained without further purification. AEROXIDE® TiO₂ P25 (Evonik) was also used as obtained.

2.2. Synthesis of studied materials

2.2.1. Preparation of mSFO

A sample of 0.722 g of SnCl₂·2H₂O was dissolved in 30 ml of deionized (DI) water. Then, pH was adjusted to 10 by adding concentrated NaOH solution, and the mixture was stirred for 1 h. Next 1.728 g of FeCl₃·6H₂O dissolved in 15 ml of DI water was added, which was followed by the rapid addition of NaOH solution (15 ml of DI water with 1.0 g NaOH). The reaction mixture was stirred for 40 min. Afterward, it was diluted subsequently with two portions of water (40 and 30 ml), while adjusting pH at 10–10.5 with NaOH solution. Finally, the mixture was transferred to a steel autoclave equipped with a PTFE insert (200 ml), heated to 200 °C for 24 h, and left to cool down to room temperature.

2.2.2. Preparation of mSFO/P25 mixture

The mSFO and P25 powders were manually ground in an agate mortar with a 2-to-1 weight ratio. The mSFO/P25 composite ratio

was selected by adjusting the mass of components to optimal photocatalytic performance (Fig. S11).

2.3. Instruments

Details about used free software and appropriate references are in the supporting information.

2.3.1. Dynamic light scattering

To determine the hydrodynamic diameter of mSFO, Malvern Zeta Sizer Nano (Malvern Instrument) was used for dynamic light scattering (DLS) measurements. The scattering angle was 173°, and the wavelength of the scattered light was 633 nm.

2.3.2. Diffuse reflectance spectroscopy

Diffuse reflectance spectra of materials were obtained with the use of Shimadzu UV-3600 UV–Vis–NIR spectrophotometer equipped with a 10 cm diameter integrating sphere. Samples mixed with BaSO₄ (white standard; proportions determined based on the reflectance signal) in the form of a tablet were placed in front of the integrating sphere. A pure BaSO₄ tablet was used as a reference. Collected spectra were recalculated to Kubelka–Munk function, and then to determine the bandgap energies of the studied samples, the Tauc transformation was applied with a baseline subtraction procedure described by Makula et al. [23].

2.3.3. Mössbauer spectroscopy

The ¹¹⁹Sn and ⁵⁷Fe Mössbauer spectra were collected at room temperature using a Polon-type spectrometer equipped with a multichannel analyzer and a linear arrangement of a ¹¹⁹Sn source in Ca¹¹⁹SnO₃ matrix (10 mCi) and ⁵⁷Co:Rh (35 mCi) source, respectively. Calibration was conducted at room temperature using a 30 μm thick α-Fe foil. The ¹¹⁹Sn isomer shifts were referenced to BaSnO₃.

2.3.4. Porosimetry

Nitrogen adsorption-desorption isotherms were obtained at 77 K with Autosorb IQ, Quantachrome Instruments. All prepared samples were thoroughly degassed at 200 °C, 50 mTorr before measurement. The specific surface areas were calculated by the multipoint Brunauer–Emmet–Teller method using the adsorption data in the relative pressure (p/p₀) range of 0.05–0.3.

2.3.5. Scanning electron microscopy

Scanning electron microscopy (SEM) analysis was performed using Tescan Vega 3 microscope with the LaB₆gun equipped with SE and BSE detectors. SiliCarbon tape was used as a substrate for imaging.

2.3.6. Transmission electron microscopy

The morphology of the mSFO sample was studied using FEI Tecnai Osiris transmission electron microscope (TEM) equipped with an X-FEG Schottky field emitter operating at the voltage of 200 kV. Before TEM observations, the mSFO sample was placed on a copper grid covered with a holey carbon film.

2.3.7. Kelvin probe

Contact potential difference (CPD) was measured using the Kelvin probe (Insytytut Fotonowy) in the air atmosphere. Surface photovoltage was measured as a variation of the surface potential on pulsed illumination (xenon lamp) vs. gold grid reference electrode (Au mesh, diameter 2.5 mm). The background baseline was subtracted from the data.

2.3.8. Photoelectrochemical measurements

The transient photocurrent measurements of the studied materials were performed in the three-electrode cell using the electrochemical analyzer equipped with the xenon lamp 150 W (Instytut Fotonowy). A thin layer of material was deposited on a transparent ITO foil (60 Ω /sq, Sigma-Aldrich) and placed as the working electrode. Simultaneously, a platinum wire and Ag/AgCl in a 3 M KCl solution, as the counter and reference electrodes, respectively, were used. The cell was filled with the acetonitrile electrolyte (1 M KNO₃ solution) and purged with Ar before (for 30 min) and during the measurement. The back side of the working electrode was illuminated by intermittent on-off illumination in the wavelength range of 330–450 nm and the applied potential range of –0.2 to 1.0 V vs. Ag/Ag⁺.

2.3.9. XPS

X-ray photoelectron spectroscopy (XPS) measurements were carried out in a Prevac photoelectron spectrometer equipped with a hemispherical VG Scienta R3000 analyzer, using the Al K _{α} (λ = 1486.6 eV) radiation. The background pressure in the analysis chamber was maintained at 10^{–8} mbar, and the pass energy was set at 100 eV. Due to the static charging of the samples, correction of the binding energy scale was adjusted by setting the C 1s peak at 284.8 eV of adventitious carbon.

2.3.10. XRD

The crystalline structure of materials was studied by powder X-ray diffractometer (XRD) Rigaku MiniFlex (PANalytical) with nickel-filtered copper K _{α} radiation (λ = 0.15406 nm, 40 kV, 15 mA). Scanning was carried out in the range of 2θ = 10–90° with 0.02° steps at a scan rate of 2° per minute.

2.3.11. XRF

The relative bulk content of tin and iron in the studied samples was determined with the use of an energy-dispersive X-ray fluorescence (XRF) spectrometer (ARL QUANT'X, Thermo Scientific). The used X-rays were in the range of 4–50 kV (1 kV step) with a beam size of 1 mm and were generated by the rhodium anode. A 3.5 mm Si(Li) drifted crystal with a Peltier cooling (~185 K) detector was used. For quantitative analysis, the calibration with a series of metallic standards and UniQuant software was used.

2.3.12. Linear sweep voltammetry

The experiments were carried out in a three-electrode setup. An Ag/Ag⁺ electrode as the reference electrode, platinum wire as the counter electrode, and platinum foil covered with the analyzed material as the working electrode were used. LiClO₄ acetonitrile electrolyte purged with Ar was used. Biologic SP-200 served as a potentiostat.

2.4. Photocatalytic tests (with GC/MS verification protocol)

Photocatalytic tests (Fig. S12) were performed in a reactor consisting of glass test tubes with silicone septum caps. Their internal volume is 13 ml; 15 mg of a photocatalyst was evenly spread on the area of 6 cm² of a glass plate, which was placed inside the test tube containing 340 μ l of DI and argon-purged water as a moisture reservoir. Next, the reactor was purged with argon and then 1.8 ml of CO₂ was injected into the reactor, while the excess pressure was released through an outlet made of a narrow needle placed shallowly into the septum. Then, the reactor was irradiated with a xenon lamp 150 W (Instytut Fotonowy) through a water filter. The absorbance of the reactor glass was measured using a UV–Vis spectrophotometer. Its material functioned as a 340 nm cutoff filter. The light intensity was measured at the sample position, and its

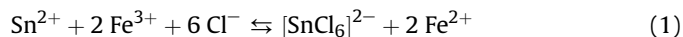
value was found to be 1 W/cm². The temperature was measured and never exceeded 36 °C. Samples of gas were taken at various times during the irradiation. Gas sampling was performed by drawing gas, using a gas-tight syringe (50 μ l) with a sample lock valve, through the septum and the samples were immediately analyzed using GC-Agilent Technologies 6890N gas chromatograph (GC) outfitted with FID and TCD. The columns used were Molsieve and HP Poraplot Q for methane, ethane, carbon monoxide, carbon dioxide, oxygen, and nitrogen analysis. The carrier gas was ultra-pure argon (impurities levels in the single ppm range) obtained from Air Products. After the irradiation process, liquid samples were drawn from the water reservoir from the bottom of the reactor and analyzed with the Total Organic Carbon analyzer (TOC-V_{cs}, Shimadzu) to establish the possibility of CO₂ reduction to the products other than CH₄ and CO in the liquid phase.

To verify that C₁ products originated from the photoreduction of CO₂, three blank experiments were carried out. One was done in the absence of a photocatalyst, the second one was done without CO₂, and the third one was done in the dark. No products were detected in these three experiments. Furthermore, photocatalytic reduction of ¹³CO₂ was carried out. Shimadzu QP2020 GC/MS was used to analyze photocatalytic reduction products to determine the source of carbon in produced CO. In this experiment, the photocatalytic reduction procedure was used as described above with the difference in carbon dioxide substrate. The N4,9 ¹³CO₂ from Sigma-Aldrich was used in these experiments. Several measurements were carried out, and the results were normalized and averaged to ensure a high degree of confidence, as peaks from carbon ion (m/z = 12 and 13) were used for analysis since in the experimental setup the column did not separate CO and N₂ which both produce ions with the same m/z = 28 (CO₂ was separated from CO on GC column).

3. Results and discussion

3.1. Synthesis and characterization of mSFO

In the synthesis of mixed Sn/Fe oxides described in the Experimental section, no organic species or solvents, which could serve as a carbon source, were applied. The developed procedure was relatively complex and extremely sensitive to pH. The relative concentration of iron and tin ions influences the formation of either the Sn-substituted Fe₃O₄ phase and/or the Fe-substituted SnO₂ phase [24]. During the synthesis, the redox reaction occurs spontaneously between Sn²⁺ and Fe³⁺ ions:



because the potential of the Sn⁴⁺/Sn²⁺ redox pair (E° = 0.151 V at 298.15 K) is lower than that of the Fe³⁺/Fe²⁺ pair (E° = 0.771 V) [25]. Therein, Sn²⁺ ions act as a sole reducing agent in the reaction environment, which allows for the further formation of the mixed valence Fe₃O₄. At the reaction pH of 10.0–10.5, metal hydroxides are formed. During the consecutive hydrothermal treatment, oxidation of the primarily formed hydroxides occurs. Essentially, tin oxides are less stable in basic solutions than iron oxides, which is a consequence of the amphoteric properties of tin. Therefore, the formation of [Sn(OH)₆]^{2–} anions is observed, reaching an equilibrium between tin in solution and solid phases [26]. The excess of Sn remaining in the solution was confirmed experimentally. Moreover, due to the lower solubility of iron oxides, the growth of SnO₂ may occur at Fe₃O₄ crystals. The resulting composite of mixed tin/iron oxides is abbreviated herein as mSFO.

The XRD analysis of mSFO (Fig. 1) reveals diffraction lines at 2θ values of 18.3°, 30.1°, 35.4°, 37.0°, 43.0°, 53.4°, 56.9°, and 62.5°,

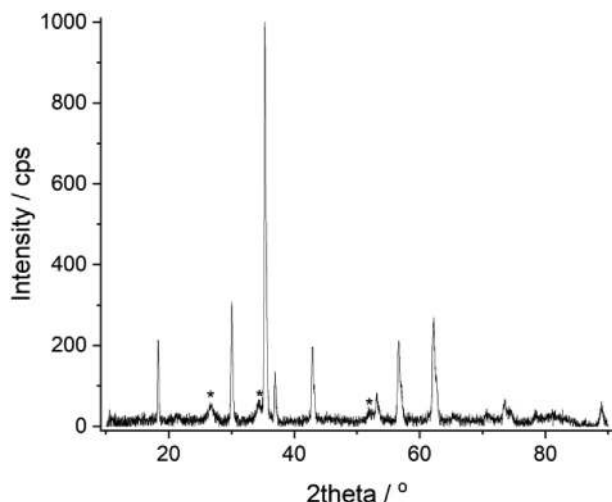


Fig. 1. X-ray diffraction of mSFO. Signals marked with asterisks (*) depict the SnO_2 phase (cassiterite), while the others are attributed to Fe_3O_4 (magnetite).

which are assigned, respectively, to (111), (220), (311), (222), (400), (422), (511), and (440) reflections of the Fe_3O_4 magnetite structure (COD entry 1,011,084). Magnetite belongs to the class of inverse spinels. Its crystallographic structure is made of two types of unit cells involving tetrahedral (A sites: Fe^{3+} surrounded by four oxygen atoms) and octahedral (B sites: Fe^{2+} or Fe^{3+} surrounded by six oxygen atoms) sites. However, octahedral sites can be occupied also by Sn^{4+} ions [27], the presence of which was confirmed by Mössbauer spectroscopy (*vide infra*). Partial substitution of other metal cations does not alter the structure of magnetite; therefore, no significant indicators of substitution can be observed in the diffractogram [28]. Herein, weak peaks marked with asterisks on the graph at 26.5° , 33.8° , and 51.7° were attributed to SnO_2 with a rutile type tetragonal structure (cassiterite; COD entry 00-152-6637) and correspond to (110), (110), and (211) planes, respectively [29]. Slight shifts of lines to larger values compared to the reference data point at Fe doping of SnO_2 [30,31]. This conclusion is also supported by Bagheri-Mohagheghi et al. who found that for SnO_2 films doped up to 7.8% with Fe^{3+} , the mean crystal size increases in (110), (101), and (211) directions [32].

To elucidate the chemical composition of mSFO, ^{57}Fe and ^{119}Sn Mössbauer spectroscopy measurements at room temperature were applied. The ^{57}Fe Mössbauer spectrum of the composite is presented in Fig. 2, and the resulting fitting parameters are collected in

Table 1. The model consisting of five components was fitted to the data and revealed a doublet and four sextets. This type of spectrum is typical for polycrystalline substituted magnetites obtained via the hydrothermal method with the $\text{Fe}_{3-x}\text{M}_x\text{O}_4$ formula [33]. The sextet marked as S1 corresponds to the tetrahedral (type A) site in magnetite, with characteristic values of isomer shift, electric quadrupole splitting, and hyperfine magnetic field, being in good agreement with the literature values for the magnetite [34]. It should be noted that substitution does not affect the parameters of this sextet. Subsequently, the superposition of remaining sextets is interpreted as the average signal from Fe^{3+} and Fe^{2+} ions at the octahedral (type B) sites, noted as $\text{Fe}^{2.5+}$ ions due to the electron hopping. The S2 sextet presents values of isomer shift, electric quadrupole splitting, and hyperfine magnetic field also characteristic of magnetite [35]. It points to the presence of Fe^{3+} ions and the absence of Sn^{4+} ions in nearby lattice sites, as the highest recorded spin density is represented by the maximum hyperfine field value [36]. The sextets marked as S3 and S4 exhibit deviations of the fitting parameters, which is a result of the impact of different numbers of nearest doping atoms [33]. Especially for S4, the disappearance of magnetic hyperfine structure (decrease of H value) and broadening of the experimental line are observed, which is consistent with the results of the magnetite doping with Sn^{4+} [37]. Concluding, the ^{57}Fe spectrum confirms a partial substitution of Fe with Sn atoms in octahedral sites in magnetite. Meanwhile, the D1 doublet originates from Fe^{3+} in the octahedral site, as the result of replacing Sn^{4+} in the tetragonal SnO_2 structure with Fe^{3+} ions [38]. The obtained parameters are comparable to those reported in the literature for Fe-doped SnO_2 [39].

Moreover, the valence state of Sn in mSFO was determined by ^{119}Sn Mössbauer spectroscopy (Fig. 2 and Table 2). The spectrum consists of only one doublet with electric quadrupole interaction, characteristic of Sn^{4+} [40]. The results show that there is no tin at other oxidation states than Sn(IV). A relatively large value of quadrupole splitting indicates the presence of defects and oxygen vacancies in the surrounding of the Sn nucleus, which are stabilized by Fe^{3+} dopants, as reported by Nomura et al. [41]. Meanwhile, the observed 0.85 mm/s experimental line width is associated with the magnetic properties of Fe_3O_4 in the composite.

The surface chemical composition of mSFO was examined using XPS. The survey spectrum of the material reveals a complex structure with the predominant presence of Sn, Fe, and O atoms (Fig. S13). To conduct the detailed analysis, deconvolution was performed (Fig. 3). The deconvoluted spectrum of mSFO in the Fe 2p region shows characteristic peaks for Fe in both oxidation states and the peak attributed to Sn 3p_{3/2} core-level electrons of Sn^{4+} at

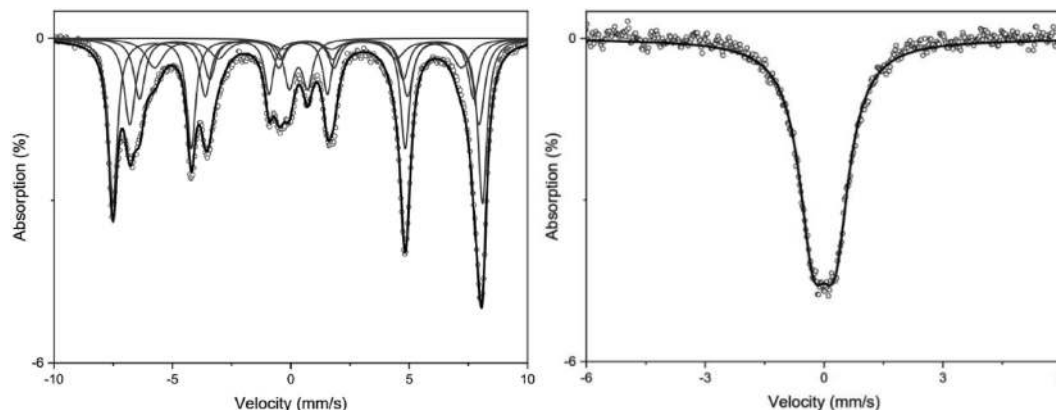


Fig. 2. ^{57}Fe Mössbauer spectrum (left) and ^{119}Sn Mössbauer spectrum (right) of mSFO at 298 K.

Table 1

Fitting parameters of the ^{57}Fe Mössbauer spectroscopic measurements of mSFO at room temperature: IS, isomer shift; QS, electric quadrupole splitting; H, hyperfine magnetic field; G, experimental line width; A, abundance of the fitted component.

Component	IS (mm/s)	QS (mm/s)	H (T)	G (mm/s)	A (%)	Oxidation state
S1	0.30	−0.02	48.6	0.45	39	(Tet) Fe^{3+}
S2	0.62	−0.07	45.9	0.55	25	(Oct) $\text{Fe}^{2.5+}$
S3	0.67	0.00	44.0	0.55	18	(Oct) $\text{Fe}^{2.5+}$
S4	0.71	0.02	40.4	0.85	12	(Oct) $\text{Fe}^{2.5+}$
D1	0.32	0.77	—	0.52	6	(Oct) Fe^{3+}

the binding energy of 716.6 eV, which is in good agreement with parameters observed for composites containing Fe_3O_4 and SnO_2 [29]. The Fe 2p region was fitted with peaks corresponding to Fe^{2+} and Fe^{3+} in Fe_3O_4 [42] and Sn $3p_{3/2}$ (Fig. 3). Moreover, the fitted peaks indicate significantly more Fe^{3+} than Fe^{2+} , which supports a magnetite structure, in agreement with XRD results. Subsequently, the high-resolution spectra in the Sn 3d region show a spin-orbit coupling doublet at the binding energy of 486.85 eV and 495.25 eV, which originate from Sn $3d_{5/2}$ and Sn $3d_{3/2}$ core-level electrons of Sn^{4+} in tetragonal SnO_2 , respectively [43]. The spin-orbit splitting separation between these peaks is equal to 8.4 eV, which is characteristic of Sn^{4+} [44]. The absence of shoulder peaks corroborates the lack of Sn^{2+} in the composite [45]. However, due to the lack of prominent binding energy shifts for different oxidation states of Sn, which is a consequence of the Madelung effects, further investigation is necessary to fully exclude the presence of Sn^{2+} [46]. It is known that the chemical environment of tin is not significantly affected by iron, which reflects in the absence of characteristic changes in the Sn 3d region [47]. Therefore, on this basis, the probability of Fe doping of SnO_2 cannot be excluded.

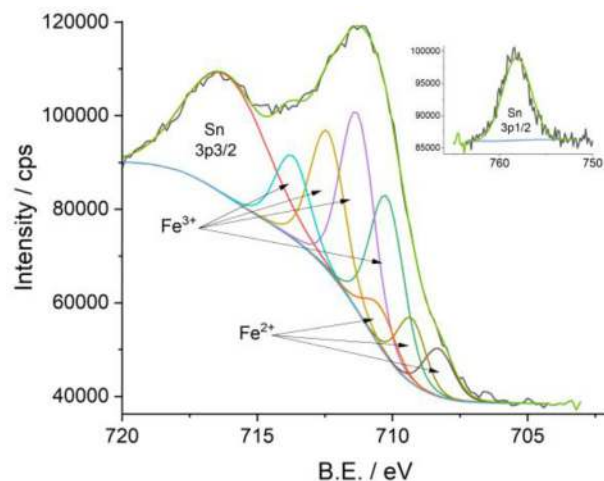
From the survey spectrum (Fig. S13), the Sn:Fe:O ratio of 1:2.5:4.3 was calculated (the organic oxygen was subtracted from the total oxygen value based on Payne et al.; Figs. S14–S15) [48]. The overabundance of oxygen atoms at the surface in form of possible hydroxyl groups or defects is well visible in the O 1s region of the XPS spectrum (Fig. S14). The XRF analysis indicates a 1:3.4 ratio of Sn to Fe in the bulk. This discrepancy in the Sn to Fe ratios, obtained from XPS and XRF, reflects the changes in the composition of the surface and bulk of the material. More Sn atoms are present at the surface than in bulk, which indicates that SnO_2 decorates the surface of Fe_3O_4 doped with Sn. The SnO_2 content is also responsible for the elevated amount of oxygen in mSFO.

The morphology of the obtained mSFO composite is presented in Fig. 4. The SEM micrograph (Fig. 4A) reveals the octahedral microstructures of Fe_3O_4 with attached nanoparticles of SnO_2 . This is corroborated by TEM microscopy (Fig. 4B), which reveals a particle size of about 0.5 μm with smaller, roughly 50 nm features. This finding is confirmed by the DLS measurements—mSFO particles with the hydrodynamic diameter of 530 nm predominate, accompanied by a small fraction of 70 nm particles (Fig. S16). The formation of similar octahedral Fe_3O_4 in solvothermal synthesis was reported by F. Ooi et al. [49]. The development of such structures is preferred at higher pH values since OH^- acts as a capping agent [50]. SnO_2 nanoparticles decorate Fe_3O_4 evenly. The relatively small size of nanoparticles could be a consequence of the restricted

Table 2

Fitting parameters of the ^{119}Sn Mössbauer spectroscopic measurements of mSFO at room temperature: IS, isomer shift; QS, electric quadrupole splitting; G, experimental line width; A, abundance of the fitted component.

Component	IS (mm/s)	QS (mm/s)	G (mm/s)	A (%)	Oxidation state
D1	0.01	0.56	0.85	100	Sn^{4+}

**Fig. 3.** XPS spectrum of Fe 2p and Sn 3p regions.

Ostwald ripening mechanism of growth under 200 °C in hydrothermal synthesis [51]. Moreover, it has been reported that doping with metals decreases the size of SnO_2 crystals [52]. Therefore, this phenomenon can also affect the size of obtained nanoparticles of SnO_2 . The described morphology of mSFO is consistent with XRD results. High-angle annular dark-field scanning TEM (HAADF-STEM) and electron energy loss spectroscopy (EELS) STEM mapping images (Fig. 4C–F) show that Fe, Sn, and O elements are distributed almost uniformly throughout the material. The micrograph reveals some small areas with a slight predominance of Fe or Sn. The described results are consistent with XRD, XPS, XRF and Mössbauer spectroscopy measurements.

3.2. Composite preparation

The mSFO sample reveals a specific surface area of 11.6 m^2/g , while the mixture of mSFO/P25 (2–1 wt ratio) shows a specific surface area of 27.1 m^2/g , which fits a weighted average of specific surface areas of mSFO and P25 (54 m^2/g [53]). This would indicate that the mixing procedure does not influence the structure of components and does not result in detrimental surface changes (Fig. S17). This is further supported by repeated N_2 adsorption measurements—after grinding mSFO alone in a mortar no change in specific surface area was observed.

3.3. Photoelectrochemical properties

The photoelectrochemical and surface photovoltage measurements were conducted to elucidate the mechanisms of photoinduced charge separation and migration. Fig. 5A presents the material photoresponse to the irradiation at 0.5 V bias. The mSFO material exhibits photoresponse in the same spectral range as P25, which probably corresponds to the light absorption by SnO_2 in the composite [54]. The photocurrent density for mSFO is an order of magnitude lower than for P25, which indicates that bare mSFO is less photoactive. However, after mixing these two materials, the photoactivity increases in the full range of probed light, which reveals the efficient electron transfer between components and the effective inhibition of the charge carriers' recombination. Bare P25 exhibits a rapid increase of photocurrent, which quickly reaches the plateau, whereas the photocurrent development in the case of mSFO/P25 is slower. This phenomenon can be assigned to electron trapping by mSFO [55].

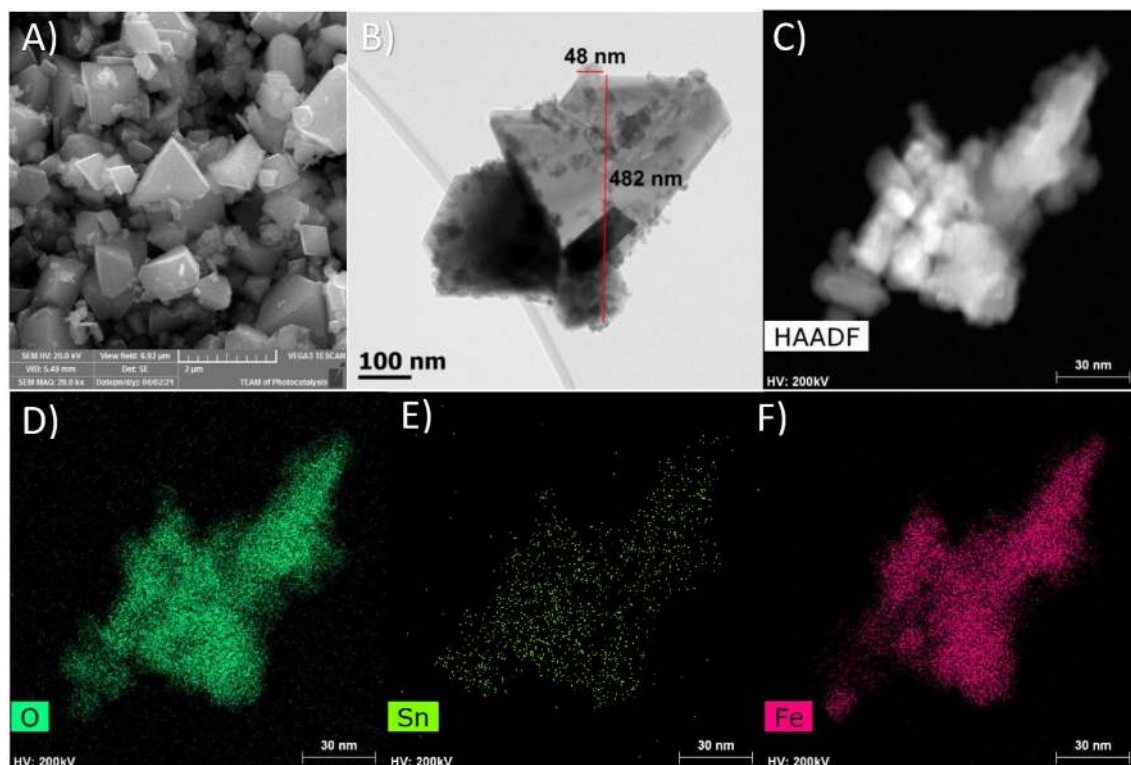


Fig. 4. SEM (A), TEM: bright field (B), HAADF (C) and EELS (D–F) micrograph of mSFO.

For a deeper understanding of the direction of charge transfer in the mSFO/P25 system, the surface photovoltage (SPV) measurements were performed (Fig. 5B). The photoresponse was measured for bare and layered materials, which were prepared by spreading thin layers of one material on top of the other one, enabling the higher light intensity to reach only the outermost one. The positive photovoltage response was observed for mSFO and P25 tested separately, which indicates that both exhibit n-type conductivity. Similar results were observed also in the case of P25 spread on top of mSFO. The photoactivity of both materials appears at 260 nm and continues to 400 and 440 nm for SFO and P25, respectively. This difference can be rationalized by the bandgap energy difference—the analysis of Tauc plots (Fig. S18) reveals the bandgap energy of mSFO equal to 3.7 or 4.1 eV for direct and indirect bandgap

estimations, respectively, while the bandgap energy of P25 is lower (ca. 3.2 eV), as reported in numerous papers [2]. In the case of mSFO covering P25, a negative CPD signal at wavelengths that are absorbed by mSFO is observed. For values higher than 400 nm, CPD is still positive due to light absorption only by P25 in that region, which is corroborated by the Tauc analysis of mSFO mixed with P25 (Fig. S19). These results reveal that mSFO could act as an efficient electron collector to enhance the separation of photo-induced electron-hole pairs, which is in good agreement with the photocurrent measurements. Fig. 6 depicts the schemes showing the direction of the electron transfer during the experiments described above. It is apparent that under irradiation, the excited electrons were transferred from P25 to mSFO. Observed interactions suggest that the mSFO-P25 system operates according to the S-scheme with

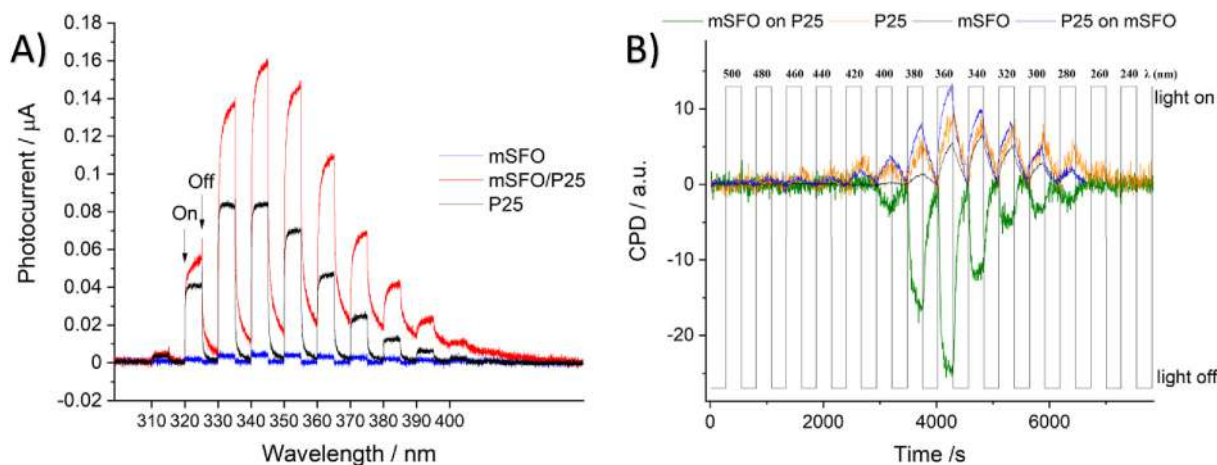


Fig. 5. Photocurrent measurements of mSFO and P25 samples (A). Surface photovoltage measurements—contact potential difference (CPD) of mSFO and P25 samples measured by Kelvin probe (B).

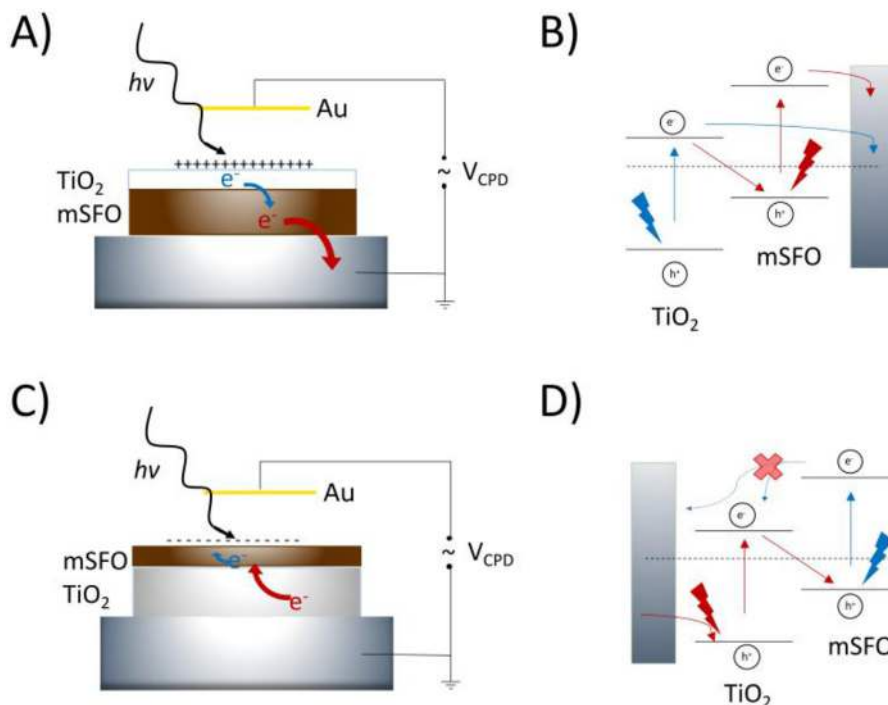


Fig. 6. Principle of SPV measurements for P25 on mSFO (A) and mSFO on P25 (C). Proposed mechanisms of charge separation in the corresponding setups (B, D).

the mSFO component offering higher energy of the CB. Therein, photoexcited TiO_2 generates the electrons, which probably recombine with VB holes of the mSFO valence band. Light absorption by mSFO results in the electron excitation to the CB, from where it may be capable of CO_2 reduction (*vide infra*). This scheme may be corroborated by linear sweep voltammetry (Fig. S110). The mSFO material can be reduced at lower potentials (higher energies) than P25, which indicates the higher energy of the CB minimum for mSFO.

3.4. Photocatalytic activity

Photocatalytic tests were carried out to check the photocatalytic activity of the mSFO and P25 composites in photocatalytic reduction of CO_2 in water (hole scavenger) saturated argon atmosphere.

Fig. 7A depicts the formation profiles of CO in the experiments with mSFO, P25, and mSFO/P25 in three consecutive runs. Expectedly, the amount of CO increases with the irradiation time. The mSFO/P25 composite photocatalyzed CO production, whereas mSFO and P25 alone did not yield any detectable CO. Specifically, the CO formation rates for mSFO/P25 in the first, second, and third runs were 0.297, 0.181, and 0.096 $\mu\text{mol/g/h}$, respectively. A decreasing photocatalytic CO_2 conversion in consecutive reaction runs may suggest poisoning of the surface by the reduction product. Fig. 7B and S111 depict GC/MS chromatograms recorded for fragmentation of $m/z = 13$ and 12. The integrated peak area for $m/z = 13$ was more than three times higher than the area for $m/z = 12$. The test was validated by carrying out an identical experiment with the use of normal $^{12}\text{CO}_2$ (Fig. S112). In this case, the ratio of the peak areas was inversed and amounted to ca. 0.5. Thus, we conclude that the

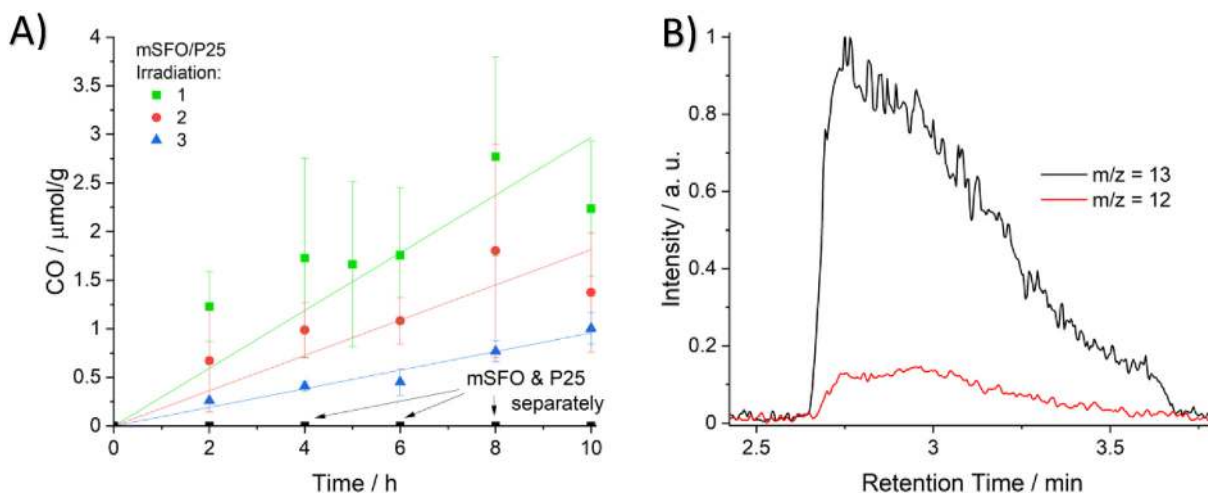


Fig. 7. Yield vs. time of photocatalytic CO_2 reduction on mSFO and P25 samples in consecutive runs (A). Gas chromatogram (GC/MS) of the gaseous sample collected after 27 h of the photocatalytic $^{13}\text{CO}_2$ reduction in the presence of mSFO/P25. The presented peak refers to the retention time characteristic of CO (B).

resultant CO in photocatalytic tests is the product of photocatalytic CO₂ reduction. Furthermore, total organic carbon (TOC) measurements carried out for the gaseous and aqueous samples (compare Fig. SI2) revealed an absence of organic carbon within the detection limit.

4. Conclusions

A new advanced mSFO/P25 system of semiconductors consisting of titanium dioxide and mSFO was designed for photocatalytic CO₂ reduction. The synthesis route for obtaining the mSFO composite, consisting of Sn(IV)-doped Fe₃O₄ microcrystals decorated with Fe(III)-doped SnO₂ nanoparticles, was proposed. The precipitation of hydroxides was followed by hydrothermal treatment. Since it is crucial to avoid any carbon sources in the synthesized material that might decompose *inter alia* to CO during CO₂ reduction experiments, the synthesis was carried out in the absence of any organic compounds. It was found that pH control constitutes a vital prerequisite for successful composite fabrication. Applied spectroscopic and microscopic studies revealed the composition, structure, and morphology of mSFO, which was further ground with titanium dioxide.

The photocatalytic performance of mSFO/P25 in the gas phase reduction of CO₂ was confirmed by GC. Carbon monoxide was the only detectable product. Its origin was verified by GC-MS analysis, which confirmed that formed ¹³CO is a product of the photocatalytic reduction of ¹³CO₂.

Linear sweep voltammetry revealed that the energy of the CB edge of mSFO is higher than the energy of the CB edge of TiO₂. In mSFO/P25, the excited electrons from TiO₂ recombine with holes photogenerated within mSFO. This electron transfer, together with CO₂ reduction taking place at mSFO and oxidation reactions occurring at TiO₂, constitute the so-called S-scheme. Correspondingly, SPV measurements confirm the photoinduced charge separation in mSFO/P25, resulting in negative and positive charging of mSFO and P25, respectively. In such a system, water oxidation with holes from VB of TiO₂ is accomplished; thus, this abundant resource can act as a hole scavenger.

Credit author statement

Kamil Urbanek: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft **Kaja Spilarewicz:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft **Jiaguo Yu:** Conceptualization, Writing – review and editing **Wojciech Macyk:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Science Center, Poland, within Sheng program (grant no. 2018/30/Q/ST5/00776).

The authors would like to thank Mr Marek Drozdek for performing XPS measurements, Dr Mariola Kądziałka-Gawel for performing Mössbauer spectrometry measurements and fitting data with models, and Dr Michał Pacia for performing TEM measurements.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mtsust.2023.100386>.

References

- [1] Z. Jiang, T. Xiao, V.L. Kuznetsov, P.P. Edwards, Turning carbon dioxide into fuel, *Phil. Trans. R. Soc. A* 368 (2010) 3343–3364, <https://doi.org/10.1098/rsta.2010.0119>.
- [2] B. Ohtani, O.O. Prieto-Mahaney, D. Li, R. Abe, What is Degussa (Evonik) P25? Crystalline composition analysis, reconstruction from isolated pure particles and photocatalytic activity test, *J. Photochem. Photobiol. C: Rev.* 216 (2010) 179–182, <https://doi.org/10.1016/j.jphotochem.2010.07.024>.
- [3] J. Schneider (Ed.), *Photocatalysis: Fundamentals and Perspectives*, Royal Society of Chemistry, Cambridge, UK, 2016.
- [4] J. Low, B. Cheng, J. Yu, Surface modification and enhanced photocatalytic CO₂ reduction performance of TiO₂: a review, *Appl. Surf. Sci.* 392 (2017) 658–686, <https://doi.org/10.1016/j.apsusc.2016.09.093>.
- [5] L. Wang, C. Bie, J. Yu, Challenges of Z-scheme photocatalytic mechanisms, *Trends Chem.* 4 (2022) 973–983, <https://doi.org/10.1016/j.trechm.2022.08.008>.
- [6] L. Wang, B. Cheng, L. Zhang, J. Yu, In situ irradiated XPS investigation on S-scheme TiO₂@ZnIn₂S₄ photocatalyst for efficient photocatalytic CO₂ reduction, *Small* 17 (2021), 2103447, <https://doi.org/10.1002/smll.202103447>.
- [7] Q. Xu, L. Zhang, B. Cheng, J. Fan, J. Yu, S-scheme heterojunction photocatalyst, *Chem* 6 (2020) 1543–1559, <https://doi.org/10.1016/j.chempr.2020.06.010>.
- [8] L. Zhang, J. Zhang, H. Yu, J. Yu, Emerging S-scheme photocatalyst, *Adv. Mater.* 34 (2022), 2107668, <https://doi.org/10.1002/adma.202107668>.
- [9] B.T. Barrocas, N. Ambrožová, K. Kočí, Photocatalytic reduction of carbon dioxide on TiO₂ heterojunction photocatalysts — a review, *Materials* 15 (2022) 967, <https://doi.org/10.3390/ma15030967>.
- [10] K. Wang, Q. Li, B. Liu, B. Cheng, W. Ho, J. Yu, Sulfur-doped g-C₃N₄ with enhanced photocatalytic CO₂-reduction performance, *Appl. Catal. B* 176–177 (2015) 44–52, <https://doi.org/10.1016/j.apcatb.2015.03.045>.
- [11] Z.-H. Wei, Y.-F. Wang, Y.-Y. Li, L. Zhang, H.-C. Yao, Z.-J. Li, Enhanced photocatalytic CO₂ reduction activity of Z-scheme CdS/BiVO₄ nanocomposite with thinner BiVO₄ nanosheets, *J. CO₂ Util.* 28 (2018) 15–25, <https://doi.org/10.1016/j.jcou.2018.09.008>.
- [12] H. Yoneyama, Photoreduction of carbon dioxide on quantized semiconductor nanoparticles in solution, *Catal. Today* 39 (1997) 169–175, [https://doi.org/10.1016/S0920-5861\(97\)00098-9](https://doi.org/10.1016/S0920-5861(97)00098-9).
- [13] T. Baran, S. Wojtyła, A. Dibeneditto, M. Aresta, W. Macyk, Photocatalytic carbon dioxide reduction at p-type copper(I) iodide, *ChemSusChem* 9 (2016) 2933–2938, <https://doi.org/10.1002/cssc.201600289>.
- [14] H. Qin, Y. He, P. Xu, D. Huang, Z. Wang, H. Wang, Z. Wang, Y. Zhao, Q. Tian, C. Wang, Spinel ferrites (MFe₂O₄): synthesis, improvement and catalytic application in environment and energy field, *Adv. Colloid Interface Sci.* 294 (2021), 102486, <https://doi.org/10.1016/j.cis.2021.102486>.
- [15] H. Han, Y. Luo, Y. Jia, N. Hasan, C. Liu, A review on SnFe₂O₄ and their composites: synthesis, properties, and emerging applications, *Prog. Nat. Sci.: Mater. Int.* (2022), <https://doi.org/10.1016/j.pnsc.2022.09.005>.
- [16] J. Guo, K. Wang, X. Wang, Photocatalytic reduction of CO₂ with H₂O vapor under visible light over Ce doped ZnFe₂O₄, *Catal. Sci. Technol.* 7 (2017) 6013–6025, <https://doi.org/10.1039/C7CY01869J>.
- [17] D. Santos-Carballal, A. Roldan, N.Y. Dzade, N.H. de Leeuw, Reactivity of CO₂ on the surfaces of magnetite (Fe₃O₄), greigite (Fe₃S₄) and mackinawite (FeS), *Phil. Trans. R. Soc. A* 376 (2018), 20170065, <https://doi.org/10.1098/rsta.2017.0065>.
- [18] Y. He, L. Zhang, M. Fan, X. Wang, M.L. Walbridge, Q. Nong, Y. Wu, L. Zhao, Z-scheme SnO₂-x/g-C₃N₄ composite as an efficient photocatalyst for dye degradation and photocatalytic CO₂ reduction, *Sol. Energy Mater. Sol. Cells* 137 (2015) 175–184, <https://doi.org/10.1016/j.solmat.2015.01.037>.
- [19] C.G. Fonstad, R.H. Rediker, Electrical properties of high-quality stannic oxide crystals, *J. Appl. Phys.* 42 (1971) 2911–2918, <https://doi.org/10.1063/1.1660648>.
- [20] J.A. Torres, G.T.S.T. Da Silva, F. Barbosa de Freitas Silva, C. Ribeiro, Experimental evidence of CO₂ photoreduction activity of SnO₂ nanoparticles, *ChemPhysChem* 21 (2020) 2392–2396, <https://doi.org/10.1002/cphc.202000786>.
- [21] A.H. Chowdhury, A. Das, S. Riyajuddin, K. Ghosh, S.M. Islam, Reduction of carbon dioxide with mesoporous SnO₂ nanoparticles as active photocatalysts under visible light in water, *Catal. Sci. Technol.* 9 (2019) 6566–6569, <https://doi.org/10.1039/C9CY01568J>.

- [22] D. Zhang, High-performance Photoelectrocatalytic Reduction of CO₂ by the hydrophilic–hydrophobic composite Cu–SnO₂/ZIF-8, *Int. J. Electrochem. Sci.* (2021), 150951, <https://doi.org/10.20964/2021.01.08>.
- [23] P. Makula, M. Pacia, W. Macyk, How to correctly determine the band gap energy of modified semiconductor photocatalysts based on UV–vis spectra, *J. Phys. Chem. Lett.* 9 (2018) 6814–6817, <https://doi.org/10.1021/acs.jpclett.8b02892>.
- [24] R. Adhikari, A.K. Das, D. Karmakar, T.V.C. Rao, J. Ghatak, Structure and magnetism of Fe-doped nanoparticles, *Phys. Rev. B* 78 (2008), 024404, <https://doi.org/10.1103/PhysRevB.78.024404>.
- [25] D. Sherwood, P. Dalby (Eds.), *Modern Thermodynamics for Chemists and Biochemists*, Oxford University Press, 2018, <https://doi.org/10.1093/oso/9780198782957.003.0002>.
- [26] L. Tan, L. Wang, Y. Wang, Hydrothermal synthesis of SnO₂ nanostructures with different morphologies and their optical properties, *J. Nanomater.* 2011 (2011), e529874, <https://doi.org/10.1155/2011/529874>.
- [27] W.-W. Wang, J.-L. Yao, Synthesis of magnetically separable Sn doped magnetite/silica core–shell structure and photocatalytic property, *Mater. Res. Bull.* 45 (2010) 710–716, <https://doi.org/10.1016/j.materresbull.2010.02.017>.
- [28] X. Liang, Y. Zhong, S. Zhu, H. He, P. Yuan, J. Zhu, Z. Jiang, The valence and site occupancy of substituting metals in magnetite spinel structure Fe_{3–x}M_xO₄ (M = Cr, Mn, Co and Ni) and their influence on thermal stability: an XANES and TG-DSC investigation, *Solid State Sci.* 15 (2013) 115–122, <https://doi.org/10.1016/j.solidstatesciences.2012.10.005>.
- [29] C. Leostean, O. Pana, M. Stefan, A. Popa, D. Toloman, M. Senila, S. Gutoiu, S. Macavei, New properties of Fe₃O₄@SnO₂ core shell nanoparticles following interface charge/spin transfer, *Appl. Surf. Sci.* 427 (2018) 192–201, <https://doi.org/10.1016/j.apsusc.2017.07.267>.
- [30] J. Hu, Y. Wang, W. Wang, Y. Xue, P. Li, K. Lian, L. Chen, W. Zhang, S. Zhuikov, Enhancement of the acetone sensing capabilities to ppb detection level by Fe-doped three-dimensional SnO₂ hierarchical microstructures fabricated via a hydrothermal method, *J. Mater. Sci.* 52 (2017) 11554–11568, <https://doi.org/10.1007/s10853-017-1319-8>.
- [31] X. Zheng, J. Cai, W. Zhao, S. Liang, Y. Zheng, Y. Cao, L. Shen, Y. Xiao, L. Jiang, Porous α -Fe₂O₃/SnO₂ nanoflower with enhanced sulfur selectivity and stability for H₂S selective oxidation, *Chin. Chem. Lett.* 32 (2021) 2143–2150, <https://doi.org/10.1016/j.ccl.2020.11.017>.
- [32] M.-M. Bagheri-Mohagheghi, N. Shahtahmasebi, M.R. Alinejad, A. Youssefi, M. Shokoh-Saremi, Fe-doped SnO₂ transparent semi-conducting thin films deposited by spray pyrolysis technique: thermoelectric and p-type conductivity properties, *Solid State Sci.* 11 (2009) 233–239, <https://doi.org/10.1016/j.solidstatesciences.2008.05.005>.
- [33] L. Diamandescu, D. Mihăilă-Tărăbășanu, V. Teodorescu, N. Popescu-Pogrión, Hydrothermal synthesis and structural characterization of some substituted magnetites, *Mater. Lett.* 37 (1998) 340–348, [https://doi.org/10.1016/S0167-577X\(98\)00117-7](https://doi.org/10.1016/S0167-577X(98)00117-7).
- [34] M.A. Shipilin, I.N. Zakharova, A.M. Shipilin, V.I. Bachurin, Mössbauer studies of magnetite nanoparticles, *J. Synth. Investig.* 8 (2014) 557–561, <https://doi.org/10.1134/S1027451014030343>.
- [35] I.S. Lyubutin, C.R. Lin, YuV. Korzhetskiy, T.V. Dmitrieva, R.K. Chiang, Mössbauer spectroscopy and magnetic properties of hematite/magnetite nanocomposites, *J. Appl. Phys.* 106 (2009), 034311, <https://doi.org/10.1063/1.3194316>.
- [36] M. Soreescu, L. Diamandescu, D. Tarabasanu-Mihaila, V.S. Teodorescu, B.H. Howard, Hydrothermal synthesis and structural characterization of (1–x) α -Fe₂O₃–xSnO₂ nanoparticles, *Solids* 65 (2004) 1021–1029, <https://doi.org/10.1016/j.jpcs.2003.10.062>.
- [37] A.M. Gismelseed, A.A. Yousif, A.D. Al-Rawas, M.E. Elzain, I. Ayub, H.M. Widatallah, F.J. Berry, in: M.F. Thomas, J.M. Williams, T.C. Gibb (Eds.), *Mössbauer Studies of the Tetravalent Doped Magnetite, Hyperfine Interactions (C)*, Springer Netherlands, Dordrecht, 2002, pp. 49–52, https://doi.org/10.1007/978-94-010-0281-3_13.
- [38] M.S. Pereira, F.A.S. Lima, T.S. Ribeiro, M.R. da Silva, R.Q. Almeida, E.B. Barros, I.F. Vasconcelos, Application of Fe-doped SnO₂ nanoparticles in organic solar cells with enhanced stability, *Opt. Mater.* 64 (2017) 548–556, <https://doi.org/10.1016/j.optmat.2017.01.023>.
- [39] C.E.R. Torres, A. Fabiana Cabrera, F.H. Sánchez, Magnetic behavior of nano-clusters of Fe-doped SnO₂, *Physica B* 389 (2007) 176–179, <https://doi.org/10.1016/j.physb.2006.07.051>.
- [40] V. Bilovol, S. Ferrari, F.D. Saccone, L.G. Pampillo, Effect of the dopant on the structural and hyperfine parameters of Sn_{0.95}M_{0.05}O₂ nanoparticles (M: V, Mn, Fe, Co), *Mater. Res. Express* 6 (2019), 0850h6, <https://doi.org/10.1088/2053-1591/ab29cc>.
- [41] K. Nomura, E. Kuzmann, C.A. Barrero, S. Stichleutner, Z. Homonnay, 119Sn Mössbauer study of sol–gel synthesized Fe_xSb_ySn_{1–x–y}O_{2–δ} powders, *Hyperfine Interact.* 184 (2008) 57–62, <https://doi.org/10.1007/s10751-008-9766-x>.
- [42] M.C. Biesinger, B.P. Payne, A.P. Grosvenor, L.W.M. Lau, A.R. Gerson, R. St. C. Smart, Resolving surface chemical states in XPS analysis of first row transition metals, oxides and hydroxides: Cr, Mn, Fe, Co and Ni, *Appl. Surf. Sci.* 257 (2011) 2717–2730, <https://doi.org/10.1016/j.apsusc.2010.10.051>.
- [43] C. Gu, W. Guan, J.-J. Shim, Z. Fang, J. Huang, Size-controlled synthesis and electrochemical performance of porous Fe₂O₃/SnO₂ nanocubes as an anode material for lithium ion batteries, *CrystEngComm* 19 (2017) 708–715, <https://doi.org/10.1039/C6CE02288J>.
- [44] J. Moulder, W. Stickle, W. Sobol, K.D. Bomben, *Handbook of X-Ray Photoelectron Spectroscopy*, Undefined, 1992. <https://www.semanticscholar.org/paper/Handbook-of-X-Ray-Photoelectron-Spectroscopy-Moulder-Stickle/6165d59e158c88267b1154c167da68bca644f4a>. (Accessed 10 August 2021).
- [45] Y. Fu, N. Sun, L. Feng, S. Wen, Y. An, J. Liu, Local structure and magnetic properties of Fe-doped SnO₂ films, *J. Alloys Compd.* 698 (2017) 863–867, <https://doi.org/10.1016/j.jallcom.2016.12.297>.
- [46] V.B.R. Boppana, R.F. Lobo, Photocatalytic degradation of organic molecules on mesoporous visible-light-active Sn(II)-doped titania, *J. Catal.* 281 (2011) 156–168, <https://doi.org/10.1016/j.jcat.2011.04.014>.
- [47] W. Ben Haj Othmen, A. Hamdi, A. Addad, B. Sieber, H. Elhouichet, S. Szunerits, R. Boukherroub, Fe-doped SnO₂ decorated reduced graphene oxide nanocomposite with enhanced visible light photocatalytic activity, *J. Photochem. Photobiol. A: Chem.* 367 (2018) 145–155, <https://doi.org/10.1016/j.jphotochem.2018.08.016>.
- [48] B.P. Payne, M.C. Biesinger, N.S. McIntyre, X-ray photoelectron spectroscopy studies of reactions on chromium metal and chromium oxide surfaces, *J. Electron. Spectrosc. Relat. Phenom.* 184 (2011) 29–37, <https://doi.org/10.1016/j.elspec.2010.12.001>.
- [49] F. Ooi, J.S. DuChene, J. Qiu, J.O. Graham, M.H. Engelhard, G. Cao, Z. Gai, W.D. Wei, A facile solvothermal synthesis of octahedral Fe₃O₄ nanoparticles, *Small* 11 (2015) 2649–2653, <https://doi.org/10.1002/sml.201401954>.
- [50] H. Fatima, D.-W. Lee, H.J. Yun, K.-S. Kim, Shape-controlled synthesis of magnetic Fe₃O₄ nanoparticles with different iron precursors and capping agents, *RSC Adv.* 8 (2018) 22917–22923, <https://doi.org/10.1039/C8RA02909A>.
- [51] K.M.Ø. Jensen, M. Christensen, P. Juhas, C. Tyrsted, E.D. Bøjesen, N. Lock, S.J.L. Billinge, B.B. Iversen, Revealing the mechanisms behind SnO₂ nanoparticle formation and growth during hydrothermal synthesis: an in situ total scattering study, *J. Am. Chem. Soc.* 134 (2012) 6785–6792, <https://doi.org/10.1021/ja300978f>.
- [52] B. Cojocar, D. Avram, V. Kessler, V. Parvulescu, G. Seisenbaeva, C. Tiseanu, Nanoscale insights into doping behavior, particle size and surface effects in trivalent metal doped SnO₂, *Sci. Rep.* 7 (2017) 9598, <https://doi.org/10.1038/s41598-017-09026-2>.
- [53] X. Wang, S.O. Pehkonen, J. Rämö, M. Väinänen, J.G. Highfield, K. Laasonen, Experimental and computational studies of nitrogen doped Degussa P25 TiO₂: application to visible-light driven photo-oxidation of As(III), *Catal. Sci. Technol.* 2 (2012) 784, <https://doi.org/10.1039/c2cy00486k>.
- [54] M. Tang, Y. Xia, D. Yang, S. Lu, X. Zhu, R. Tang, W. Zhang, Ag decoration and SnO₂ coupling modified anatase/rutile mixed crystal TiO₂ composite photocatalyst for enhancement of photocatalytic degradation towards tetracycline hydrochloride, *Nanomaterials* 12 (2022) 873, <https://doi.org/10.3390/nano12050873>.
- [55] L.M. Peter, Dynamic aspects of semiconductor photoelectrochemistry, *Chem. Rev.* 90 (1990) 753–769, <https://doi.org/10.1021/cr00103a005>.